Timing Search[†]

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Abstract

Why do industries fluctuate at different frequencies? I study a timing search model in which entrants choose dates and rapid adjustment congests a shared input. In equilibrium, entrants are indifferent across entry dates, so profit net of the congestion cost just covers the entry cost, pinning the speed of adjustment at each scale of the industry. The model admits multiple equilibria: the industry may rest at either break-even scale, cycle between them, or wither away. I focus on the cyclical equilibrium, the unique dynamically robust equilibrium within the full-support class. This cycle repeats at frequency $\omega = \sqrt{\varphi/\mu}$, set by the industry-level profit curvature φ and the timing friction μ. In U.S. manufacturing, recovery times rise with recession depth at the predicted exponent of one-half, and annual establishment-count frequencies rise with concentration. Finally, I derive an analytical state-dependent Pigouvian entry wedge and study its policy implications.

Keywords: Timing search, equilibrium multiplicity, industry frequency, entry timing, Pigouvian taxation.

JEL codes: D83, E32, L16, E22.

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1 Introduction

Why do industries fluctuate at different frequencies? U.S. production and establishment entry exhibit substantial cross-industry dispersion in their dominant cycle frequency despite common aggregate conditions. Standard business-cycle frameworks usually tie cycle dynamics to a common shock process (Kydland and Prescott, 1982) or a common policy rule. This paper answers with industry-specific market structure.

I propose a *timing search* equilibrium, a temporal version of directed search (Moen, 1997; Burdett and Mortensen, 1998; Wright et al., 2021): entrants choose their entry dates under a dynamic entry friction. Timing search extends directed search from contemporaneous market choice to intertemporal market choice. Dates play the role of submarkets, but time is a special index. A submarket indexed by a posted wage bears no physical relation to its neighbors; dates do, because time runs one way. Each date inherits the industry left by earlier entry, and an entrant can wait but cannot move backward. The friction arises from this contiguity: changing the inherited level quickly between neighboring dates pushes entrants through the same entry pipeline of permits, specialist labor, equipment, and financing all at once. Congestion therefore operates across neighboring dates rather than at a single date, extending the static within-submarket congestion of directed search to a dynamic setting. The framework thereby connects directed search with the literature on frictional intertemporal adjustment, in which convex and fixed adjustment costs shape firm dynamics (Abel and Eberly, 1994; Caballero and Engel, 1999; Cooper and Haltiwanger, 2006) and staggered or state-dependent adjustment shapes pricing (Calvo, 1983; Golosov and Lucas, 2007).

In equilibrium, no entry date can offer a strictly better payoff than another; if one did, entrants would move toward it until the advantage disappeared. This date-by-date indifference is the temporal counterpart of the equal-value condition in directed search, and an intertemporal-arbitrage condition whose closest formal precedent is Hotelling (1931). It is also the standard free-entry condition of endogenous-entry models (Hopenhayn, 1992; Melitz, 2003; Bilbiie et al., 2012), holding entry value at entry cost at every active date. What is new is that rapid adjustment lowers effective profit, so the same condition ties the industry's speed to its scale rather than pinning the mass of entrants.

The model has a single predetermined aggregate state, the mass of active establishments n_t. I assume each industry has a maximal-profit scale: per-establishment profit is single-peaked in n_t at n_p. When the industry adjusts quickly, entrants congest the shared input, so effective profit also falls with the speed of net adjustment $\dot{n}_t$, symmetrically for expansion and contraction. The congestion an entrant faces at date t therefore depends on the pace of adjustment $\dot{n}_t$, not on how many firms are already active n_t. In equilibrium, every date must remain equally attractive, so high profit must be offset by congestion. The industry adjusts fastest at the maximal-profit scale and slows to a halt where profit falls to the entry cost. It cannot rest at the peak; it sweeps through and reverses at each break-even scale. The result is the endogenous cycle around the peak n_p. The semiconductor industry illustrates this mechanism. When chips are profitable, firms race to build fabrication plants, but every entrant draws on the same scarce inputs: cleanroom construction, lithography equipment, process engineers. Capacity keeps arriving after profits have peaked, the rush overshoots into a glut, and the correction overshoots into scarcity. The pattern is known in the industry as the silicon cycle. The industry's own profit landscape and entry bottleneck set how fast this repeats.

The contributions are threefold. First, I characterize the endogenous cycle that arises from timing search and show that its frequency is set by industry-specific primitives (Theorem 1, Section 3). Under the log-quadratic benchmark the industry follows a harmonic path with closed-form angular frequency

$$\omega = \sqrt{\varphi/\mu}, \tag{1}$$

where φ is the curvature of profits around the maximal-profit scale and μ is the timing friction parameter. The cycle coexists with rest at a break-even scale and with withering, but it is the unique dynamically robust full-support equilibrium (Proposition 2) and the one relevant to the observed dispersion in frequencies.

Second, I take the model to U.S. manufacturing (Section 4). The main test is the equilibrium response to shocks: recovery times rise with recession depth at the parameter-free exponent of one-half that phase displacement predicts. This rejects both depth independence and constant-rate recovery (Section 4.3). The concentration ranking uses one spectral rule and an Ellison–Glaeser index built from public County Business Patterns data. The frequency of the model's state variable, the establishment count, rises with concentration, and the slope is distinguishable from zero. Production frequencies share the sign imprecisely, and the estimated peaks are precise only to a factor of about 1.6, so the cross-industry ordering is broad rather than sharp. Two historical windows make the object concrete: manufacturing investment repeated a five-year rhythm for four decades, and oil investment after 1979 a 3.25-year rhythm for a quarter century (Section 4.4).

Third, private date choice equates an entrant's own value across dates but ignores the externalities it imposes on incumbents: the scale effect on their profits and the congestion effect during adjustment. I derive the state-dependent Pigouvian entry wedge that decentralizes an interior planner allocation (Proposition 5, Section 5). The simpler flat tax that collapses the private locus to $(n_p, 0)$ permits extinction from any inherited scale, so policy inherits the same multiplicity as the positive model.

Related literature This paper builds on four literatures.

Endogenous cycles. The plucking tradition interprets recessions as departures below a ceiling and emphasizes deeper-recession–faster-recovery patterns (Friedman, 1993; Bordo and Haubrich, 2017; Dupraz et al., 2019). That tradition takes the size of the departure as given and studies the return; free entry here pins the size itself, so an industry has a single orbit rather than a family indexed by the size of a disturbance. Beaudry et al. (2020, 2024) obtain continuing stochastic motion near a Hopf bifurcation. Timing search instead classifies an equilibrium correspondence organized by a conserved locus, which also contains two endpoint rest paths.

A lag-driven tradition generates industry-specific frequencies from production delays: cobweb dynamics tie the period to the production lag (Ezekiel, 1938; Nerlove, 1958), and Rosen et al. (1994) generate cattle cycles from gestation and holding lags under rational expectations. In the modern entry pair, Kalouptsidi (2014) quantifies time to build and shipyard congestion in bulk shipping, and Greenwood and Hanson (2015) generate ship-price waves from extrapolative entry; the shipyard queue is the closest quantitative counterpart of the entry pipeline here. In each case the frequency is inherited from a delay: cobweb cycles run at twice the lag, and time-to-build cycles at the build horizon. Timing search delivers the frequency from the profit landscape and the congestion friction, with rational entrants indifferent across dates and no delay structure.

An interaction-based tradition generates aggregate fluctuations by coordinating discrete adjustments (Nirei, 2006, 2015; Nirei and Scheinkman, 2024). Timing search instead uses congestion across neighboring dates: complementarity clusters adjustment into spikes, whereas congestion spreads entry over time. Shleifer (1986) also obtains cycles from adjustment timing, but through demand complementarity that synchronizes firms into one economy-wide cycle.

Models of animal spirits and self-fulfilling prophecies generate fluctuations from indeterminacy and beliefs (Farmer and Guo, 1994; Farmer, 1999); earlier cycle precedents include Grandmont (1985), Azariadis (1981), and Benhabib and Farmer (1994). Closest on timing, Jovanovic (2009) studies the option value of investment dates. The present paper instead studies equal values across all active dates and classifies the resulting paths.

Entry timing. The entry margin is strongly procyclical and propagates aggregate shocks (Lee and Mukoyama, 2015; Clementi and Palazzo, 2016). A complementary literature makes *when* to enter a choice. The option to delay changes the opportunity cost of entry (Vardishvili, 2026); time to build propagates business cycles (Cook, 2001); and sunk entry with a lag transmits shocks through product variety (Bilbiie et al., 2012). Timing search shares the timing margin but locates its contribution in equal values across active dates and the associated equilibrium classification. These environments close entry with the free-entry condition, the value of entering equal to its cost at every date with positive entry (Hopenhayn, 1992; Melitz, 2003; Bilbiie et al., 2012). Timing search keeps that condition. When profit depends on scale alone, date-by-date free entry holds the industry at a break-even scale. When effective profit also falls with the pace of adjustment, the same condition traces a locus in scale and speed whose still points are the standard zero-profit outcome and whose interior is the endogenous cycle.

Search and adjustment frictions. Timing choice is in the spirit of Stigler (1961), McCall (1970), and especially the equal-value condition in Burdett and Mortensen (1998). Dates, however, are inherited states rather than market-clearing submarkets. The reduced-form pipeline friction connects to matching and adjustment-cost approaches (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000; Caballero and Hammour, 1994), but no matching market-clearing equation is imposed.

Industry dynamics. The model isolates timing within an endogenous-entry environment and is complementary to the heterogeneity and selection mechanisms of Hopenhayn (1992); Hopenhayn and Rogerson (1993). Startup cohorts inherit conditions at birth in Sedláček and Sterk (2017) and Pugsley et al. (2021); here each date inherits the stock created by earlier cohorts. Work on agglomeration motivates the single-peaked profit landscape (Krugman, 1991; Duranton and Puga, 2004; Rosenthal and Strange, 2004; Syverson, 2004).

2 Model

The environment has three elements: a single-peaked profit landscape, a timing friction, and a linear-utility household. A matching calculation provides an illustrative calibration of the friction but is not part of equilibrium market clearing.

2.1 Agents

The economy has two types of agents. A representative household consumes and supplies labor. A finite pool of potential entrepreneurs is either outside or operating a firm; while outside, an entrepreneur can wait before entering. The household has linear utility in consumption and linear disutility from labor, $U = \int_0^\infty e^{-\rho t}\big[C_t - \bar{w}\, L_t\big]\, dt$. The static labor-supply condition then pins $w = \bar{w}$, and labor is supplied elastically.[1] I impose the common deterministic discount rate $r = \rho > 0$, so private and planner discounting coincide. Three ingredients define the entry side of the environment.

[1]This quasilinear specification is in the spirit of Greenwood et al. (1988) preferences in eliminating the wealth effect on labor supply.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of potential entrepreneurs is either outside *or* active*; an outside entrepreneur may wait costlessly and choose a later entry date. The outside option is zero, and participation into timing choice is competitive whenever the pool is not exhausted. Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = zl_i^\gamma$, $\gamma \in (0, 1)$. Each entry uses $\kappa > 0$ units of real entry resources, paid at the moment of entry. Firms exit at Poisson rate $\delta > 0$, returning their entrepreneur to the outside pool.*

Let n_t denote the mass of active production units, with $n_t < M$ on the full-support paths studied below. The outside mass supplies additional participants if entry value exceeds its real resource cost; interior participation therefore gives zero net surplus while the pool remains slack. The household ultimately owns the operating claims and bears the entry-resource outlay κ. Waiting has no flow payoff. I interpret n as active *establishments* rather than legal firms, matching the aggregate-entry interpretation below.

2.2 Scale efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms crowd the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$, with productivity z and labor l_i. Because firms reoptimize labor against the scale effect, optimized profit raises it to the power $1/(1-\gamma)$: $\pi(n) = \bar{\pi}(\bar{w})\, m(n)^{1/(1-\gamma)}$, where $\bar{\pi}(\bar{w})$ is the standard per-firm profit at the fixed wage $\bar{w}$ of Section 2.1, earned absent any scale effect.[2]

The classification requires only a single-peaked profit curve.

Assumption 2 (Single-peaked profits)**.**
The profit function $\pi : [0, M) \to \mathbb{R}_{++}$ is continuous at zero and C^2 on $(0, M)$. It has a unique peak n_p: $\pi'(n) > 0$ for $n < n_p$ and $\pi'(n) < 0$ for $n > n_p$. Let $\bar{y} \equiv (r+\delta)\kappa < \pi(n_p)$, and suppose $\pi(n) = \bar{y}$ has two solutions, the break-even scales n_- and n_+, with $0 < n_- < n_p < n_+ < M$.

I call n_p the industry's maximal-profit scale. Single-peakedness allows productive thickness at low scale and crowding at high scale. It also makes the two break-even crossings transverse, a condition that ensures finite travel time at the turning points. The productivity-density evidence in Syverson (2004) motivates non-monotone scale effects, but does not identify the shape of this aggregate profit curve.

[2] The static labor choice gives $\pi(n) \propto [zm(n)]^{1/(1-\gamma)}$.

The log-quadratic case gives the benchmark:

$$\pi(n) = \pi_0 \exp\left(-\frac{\varphi}{2}(n - n_p)^2\right), \qquad \pi_0 > 0. \tag{2}$$

It arises from $m(n) = \exp[-(1-\gamma)\varphi(n-n_p)^2/2]$ in the production structure above; Online Appendix A.1 gives an illustrative foundation. The normalization makes φ the curvature of log profit after labor reoptimization. Its semi-elasticity is linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{3}$$

High φ means a narrow profit peak; low φ a broad one. This linearity delivers exact harmonic motion as a corollary of the general classification.

2.3 Timing friction

The benchmark takes the timing friction as reduced form. It penalizes rapid *net industry adjustment*: expansion strains permitting and construction, while contraction strains resale, scrapping, and reorganization. The state derivative matters because it records how rapidly the inherited industry is being changed.

Assumption 3 (Timing friction)**.**
The multiplicative timing friction takes the log-quadratic form

$$\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}, \qquad \mu > 0,$$

where μ is the friction parameter.

The functional form makes effective profit fall with the squared adjustment rate, with μ a free parameter. Any smooth symmetric friction has the local expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$. Symmetry is essential for the exact closed form. Online Appendix A.2–A.3 gives a discrete-time motivation and a matching-based scale for μ; neither is imposed as market clearing.

The net-flow specification is restrictive. With large simultaneous entry and exit, net adjustment understates traffic through a gross-entry pipeline. The model instead treats both expansion and contraction as costly changes in an inherited industry state. A separate-pipeline model is an extension, not an alternative interpretation of the present state variable.

2.4 Combined structure

Effective profit is

$$y_t = \pi(n_t) \exp\left(-\frac{\mu}{2}\dot{n}_t^2\right). \tag{4}$$

The general theorem uses only this expression and Assumption 2. Under (2), it becomes a bivariate log-quadratic kernel centered at $(n_p, 0)$.

2.5 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning the effective profit $y_t = \pi(n_t)\,\xi(\dot{n}_t)$ of Section 2.4 per unit time, discounted at rate $r > 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, y_t \,\mathrm{d}t. \tag{5}$$

$V(\tau)$ is the value of a single entry spell: operating from τ until stochastic exit. Re-entry after exit is a fresh participation decision. Under timing indifference, the single-spell formulation is therefore without loss for the full-support paths studied here. The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n_t\}$ as given. An agent who does not enter at date t can wait, with continuation value

$$V_t^{\text{search}} = \max\left\{0,\ \sup_{\tau \geq t} e^{-r(\tau-t)}\big[V(\tau) - \kappa\big]\right\}, \tag{6}$$

where κ is the per-firm real entry-resource cost from Assumption 1, paid at the moment of entry.

2.6 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition n_0, it determines the entire path of the economy through the aggregate consistency condition $\dot{n}_t = e_t - \delta n_t$.

Definition 1 (Full-support timing-indifference equilibrium)**.**
A full-support timing-indifference equilibrium *is a path $\{n_t\}_{t\geq 0}$ and an entry policy $e(\cdot)$ such that:*

(i) ***Individual optimality****: Each participant maximizes discounted continuation value, comparing immediate entry against waiting and entering later, with waiting value V_t^{search} of (6).*

(ii) ***Timing indifference****: participation and entry are positive at every date, the outside pool is slack, and competitive participation gives $V_t^{search} = 0$ and $V(\tau) = \kappa$ for every τ.*

(iii) ***Aggregate consistency****: $\dot{n}_t = e_t - \delta\, n_t$, where $e_t \geq 0$ and e is continuous on $(0, \infty)$.*

(iv) ***Feasibility****: $n_t < M$ for all t.*

Continuity of the entry flow The continuity in condition (iii) binds away from a path's initial date: date zero, the date an unanticipated shock re-initializes the economy from the inherited scale, or the date entry resumes after a no-entry phase. At such a date the entry flow may jump, because the scale is the only inherited state and the speed is determined by the current flow.

Interiority as a maintained restriction Ordinary participation gives complementary slackness: positive participation at a date implies $V_t = \kappa$, while zero participation permits $V_t \leq \kappa$. Definition 1 studies the stronger case in which every date is active and the outside pool remains nonempty, so equality holds everywhere. The amplitude bound below verifies $e_t > 0$ and $n_t < M$ for the harmonic paths and stationary endpoints. It does not show that every equilibrium must have full support. Proposition 1 characterizes the consequence of this maintained interiority restriction.

Commitment versus dynamic optimization The formulation in (5)–(6) is a commitment problem: agents choose τ taking the future path $\{n_t\}$ as given. Timing indifference makes it equivalent to the dynamic problem in which agents observe $(n_t, \dot{n}_t)$ and decide whether to enter at each instant: since $V(\tau) = \kappa$ for all τ in equilibrium, an agent free to re-optimize finds no profitable deviation and chooses not to re-time.

2.7 Timing indifference

The next proposition turns the no-retiming idea into an equation. If the value of entry is the same at every date, then a small delay cannot change that value. Free entry, imposed at every date, therefore reappears in flow form: the flow payoff must equal the annuity value of the entry cost. I call this the *flow condition.*

Proposition 1 (Free entry in flow form)**.**
Full-support timing indifference (Definition 1(ii)), $V(\tau) = \kappa$ for all τ, implies the flow free-entry condition almost everywhere:

$$\pi(n_t)\,\xi(\dot{n}_t) = (r + \delta)\kappa \qquad \textit{for a.e. } t. \tag{7}$$

In a classical full-support equilibrium it holds at every date.

Proof. By Definition 1(ii), entry occurs at every date and $V(\tau) = \kappa$ for all τ. The bounded flow payoff makes $V(\tau)$ Lipschitz and hence differentiable almost everywhere.[3] Differentiating (5):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Since $V \equiv \kappa$, its derivative is zero and the displayed condition follows almost everywhere. In a classical full-support equilibrium, n and $\dot{n}$ are continuous, so it extends to every date. Because the symmetric friction treats $+|\dot{n}|$ and $-|\dot{n}|$ identically, the flow condition determines speed, not the initial direction. ■

Proposition 1 is a deterministic, pathwise result for a constant entry-resource cost. A conditional slow-moving perturbation is reported separately in Online Appendix B.

Comparison with a standard free-entry condition Conditional on the maintained full-support restriction, Proposition 1 imposes nothing beyond free entry. In a stationary environment the condition is a single equality, $V = \kappa$, pinning the mass of entrants (e.g., Hopenhayn, 1992; Melitz, 2003); in dynamic entry models it holds at every date with positive entry (e.g., Bilbiie et al., 2012), as in Definition 1(ii). The differentiation step is likewise standard: a marginal delay cannot raise an entrant's value, so $\mathrm{d}V/\mathrm{d}\tau = 0$ and the level equality becomes the flow condition (7). What distinguishes timing search is the payoff the condition acts on. If profit depended on scale alone ($\xi \equiv 1$), the flow condition would read $\pi(n_t) = (r+\delta)\kappa$ at every date, and free entry would hold the industry still at a break-even scale; the rest equilibria of Theorem 1 below are that outcome. Because effective profit also falls with the speed of adjustment, the break-even point opens into a locus: $\pi(n)\xi(\dot{n})$ stays at the annuity value $(r+\delta)\kappa$ along a curve in $(n, \dot{n})$, trading scale against speed. The classical condition survives at the still points: at $\dot{n} = 0$ the friction is inactive, $\xi(0) = 1$, and the flow condition reduces to the zero-profit equality whose two roots are the break-even scales $n_{\pm}$ of Assumption 2. The cycle is therefore not the product of a stronger equilibrium condition; it is what free entry permits once rapid adjustment is costly.

3 Equilibrium analysis

3.1 The equilibria

The full-support definition is stronger than ordinary free-entry complementarity. At an inactive date the ordinary condition permits $V(t) \le \kappa$ and $e_t = 0$; equality is required only

[3] The first-order condition holds almost everywhere and extends to every date once the integrand has a continuous version.

when entry is positive. That distinction adds a withering equilibrium, separate from any question of how a moving path continues.

Among the full-support equilibria, timing indifference gives another source of multiplicity. Write the break-even flow payoff as $\bar{y} = (r + \delta)\kappa$ and define

$$h(n) \equiv \ln \frac{\pi(n)}{\bar{y}}.$$

Taking logs in (7) gives the constant-payoff locus

$$\frac{\mu}{2}\dot{n}_t^2 = h(n_t). \tag{8}$$

Assumption 2 implies that h is positive between n_- and n_+, zero at those endpoints, and crosses zero with a nonzero slope. The locus confines the state to $[n_-, n_+]$ and fixes the absolute speed at every interior scale. Its endpoints are stationary full-support equilibria; the same locus also contains cyclical candidate paths.

Effective profit must be the same at every active date, but effective profit falls both with distance from the profit peak and with the speed of adjustment. An industry far from its peak must therefore be moving slowly, and an industry near its peak must be moving fast. This requirement pins speed to scale.

The locus does not determine which way the industry travels along it, or whether it travels at all. The log-quadratic benchmark settles that next, and the answer is three kinds of equilibrium.

Under the log-quadratic landscape (2), write

$$b \equiv \ln \frac{\pi_0}{\bar{y}} > 0, \qquad A \equiv \sqrt{\frac{2b}{\varphi}},$$

so that the break-even scales are $n_- = n_p - A$ and $n_+ = n_p + A$ and the locus (8) becomes the ellipse

$$\frac{\varphi}{2}\,(n_t - n_p)^2 + \frac{\mu}{2}\,\dot{n}_t^2 = b. \tag{9}$$

The analysis focuses on equilibria with a smooth aggregate entry schedule: $e \in C^1(\mathbb{R}_+)$, equivalently $n \in C^2(\mathbb{R}_+)$ since $e_t = \dot{n}_t + \delta n_t$. Call these *classical*. The restriction rules out a kink in the density of chosen entry dates; Proposition 2 shows it is inessential for selecting the cycle. Direction is settled already by the continuity in Definition 1(iii): the sign of $\dot{n}$ is transmitted continuously and can reverse only at the still points, so direction is free only where a path begins, at date zero or at an unanticipated shock. Classical smoothness adds that the continuation law forces the turn there rather than a pause.

State and recursive representation The predetermined aggregate state is the scale alone: the speed $\dot{n}_t = e_t - \delta n_t$ is determined by the current entry flow, so it is a jump variable rather than a state, and $\ddot{n}_t$ is its implication rather than a coordinate. A recursive equilibrium on n alone, however, cannot carry a cycle, since every interior scale is visited twice per period, once expanding and once contracting. The recursive representation therefore pairs the scale with the direction of travel: entrants take an aggregate policy $e = E(n, d)$ with $d \in \{+, -\}$, where $E(n, d) = \delta n \pm \sqrt{2h(n)/\mu}$ on the cycle and d reverses at the break-even scales. The direction is payoff-irrelevant, because the friction is symmetric. It is the coordination variable that continuity of the entry flow transmits along a path, the recursive counterpart of the phase in Theorem 1. In this representation the equilibrium value function is the constant $V \equiv \kappa$, so the fixed point is trivial in values, and the equilibrium content sits entirely in the consistency of the policy with the flow condition.

Theorem 1 (Classification of equilibria)**.**
Under Assumptions 1–3 and (2), the model admits three kinds of equilibria.

(i) ***Rest.*** *$n_t \equiv n_p - A$ and $n_t \equiv n_p + A$.*

(ii) ***Cycle.*** *For any phase ϕ,*

$$n_t = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}, \qquad T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{\mu}{\varphi}}, \tag{10}$$

provided

$$A < \min\left\{\frac{\delta n_p}{\sqrt{\omega^2 + \delta^2}},\ M - n_p\right\}. \tag{11}$$

(iii) ***Withering.*** *$e_t = 0$ and $n_t = n_0 e^{-\delta t}$, from any inherited n_0, whenever $\bar{y} \geq \bar{y}_c \equiv \sup_{m\in[0,M)} \pi(m)\,\xi(-\delta m)$. Since $\bar{y}_c < \pi(n_p)$, the admissible interval $[\bar{y}_c, \pi(n_p))$ for the break-even level $\bar{y}$ is nonempty.*

Proof. Let $U = \{t \geq 0 : \dot{n}_t \neq 0\}$. Differentiating (9) on U and cancelling $\dot{n}_t$ gives

$$\mu\ddot{n}_t + \varphi\,(n_t - n_p) = 0. \tag{12}$$

If U is empty, (9) leaves only $n_t \equiv n_p \pm A$. If U is nonempty, let t_0 be a boundary point of U: continuity and (9) give $\dot{n}_{t_0} = 0$ and $n_{t_0} = n_p \pm A$, and taking limits along U with $n \in C^2$ gives $\ddot{n}_{t_0} = \mp\varphi A/\mu \neq 0$. So t_0 is an isolated zero of $\dot{n}$ and a relative neighborhood of it lies in $\overline{U}$; the closure $\overline{U}$ is then open as well as closed in the connected set $\mathbb{R}_+$, so $\overline{U} = \mathbb{R}_+$. Hence U is dense and continuity extends (12) to every date. Its solutions are (10), with the amplitude fixed at A by (9) and the period $2\pi/\omega$.

For feasibility, $e_t = \delta n_p + A[\delta\cos(\omega t + \phi) - \omega\sin(\omega t + \phi)]$, whose minimum is positive exactly under the first bound in (11); the second keeps the outside pool slack. Each rest path has $e_t = \delta n_\pm > 0$. Along (i) and (ii) effective profit equals $\bar{y}$, so $V(\tau) = \kappa$ at every date.

For (iii), the flow payoff on the withering path is $\pi(n_t)\xi(-\delta n_t) \leq \bar{y}_c \leq \bar{y} = (r+\delta)\kappa$ at every date, so $V(\tau) = \int_\tau^\infty e^{-(r+\delta)(s-\tau)}\pi(n_s)\xi(-\delta n_s)\mathrm{d}s \leq \kappa$ for every τ and complementary slackness makes $e_\tau = 0$ optimal throughout. That $\bar{y}_c < \pi(n_p)$ holds because the product equals $\pi(n_p)$ only if $\pi(m) = \pi(n_p)$ and $\xi(-\delta m) = 1$, i.e. $m = n_p$ and $m = 0$ together, which $n_p > 0$ forbids: on the compact $[0, n_p]$ the continuous product attains a maximum strictly below $\pi(n_p)$, and on $[n_p, M)$ it is at most $\pi(n_p)\xi(-\delta n_p) < \pi(n_p)$. ■

Cases (i) and (ii) exhaust the classical full-support equilibria: every nonstationary one is (10), and the only stationary ones are the two rest paths. Case (iii) drops full support, since no date is active. Figure 1a draws the locus and its equilibria. Proposition 2 below selects case (ii) as the dynamically robust equilibrium, and the analysis focuses on it. Online Appendix B.6 verifies each condition of Definition 1 along the harmonic paths.

Each kind of equilibrium is coherent for a different reason. On the cycle, entrants keep arriving through the expansion because profits are still above the entry cost, and they keep arriving past the peak because stopping would leave the speed too low for the date-by-date indifference to hold. At a break-even scale the industry is already so far from its peak that profits exactly cover entry with no adjustment at all, so replacement entry alone sustains it. An industry can wither through a standard coordination failure: entering an industry that nobody else enters means joining a shrinking market, which is not worth the entry cost, and the expectation is self-confirming. Even an industry at its profit peak can wither in this way.

Intuition The frequency has a mechanical reading. Equation (12) is the equation of a mass on a spring, $\mu\ddot{n} = -\varphi\,(n - n_p)$: profit curvature φ is the *restoring force*, since a steeper profit peak punishes deviations harder and pulls the industry back faster, while the timing friction μ is the *inertia*, since a stiffer pipeline makes the industry slower to turn. A stiff spring with a light mass oscillates quickly; a slack spring with a heavy mass oscillates slowly. The industry does the same, at $\omega = \sqrt{\varphi/\mu}$, exactly as a mass-spring system with stiffness k and mass m runs at $\sqrt{k/m}$. Both primitives are needed: with no scale externality the spring has no stiffness, and with no friction it has no inertia.

The analogy also explains why the frequency is free of the orbit's size. The flow condition fixes maximal speed at the profit peak and slower motion near the break-even scales, and in the log-quadratic case speed and distance scale together: a deeper orbit travels a longer way, but proportionally faster, and the two effects cancel. A cycle therefore takes the same time around whether it is small or large. The same cancellation makes the frequency free of productivity. A multiplicative increase in productivity lifts the whole profit curve, which enlarges the orbit at a fixed raw entry cost, but it leaves the *curvature* of log profit untouched (the spring is no stiffer for being higher off the ground), so the frequency does not move.

The full-support paths on the ellipse (9) are the harmonic cycle (10), the two rest paths $n_t \equiv n_\pm$, and paths that pause at a break-even scale for part of their course. Robustness is tested with a short disturbance to the entry flow that ends at date s. While it runs, the disturbance moves the flow and with it the speed; what it leaves behind is its terminal scale, some interior $n_s^\varepsilon \in (n_-, n_+)$ with $|n_s^\varepsilon - n_s| \le \varepsilon$. On returning to the full-support class, free entry pins the speed, with the direction d_s inherited from the pre-disturbance path: $\dot n_s^\varepsilon = d_s\sqrt{2h(n_s^\varepsilon)/\mu}$. A continuation is a classical equilibrium path of the same environment from this state; from a rest or a pausing path, no direction is inherited while the industry is stationary, so an inward continuation in either direction is admissible. A full-support equilibrium $\{n_t\}$ is *dynamically robust* if $\sup_{t\ge s}|n_t^\varepsilon - n_t| \to 0$ as $\varepsilon \to 0$, for every date s and every such continuation. A displacement that leaves the band leaves the full-support class, and Proposition 3 below covers it: from above, entry stops and the scale decays back to the band edge; from below, the industry withers.

Proposition 2 (Dynamic robustness selects the cycle)**.**
The harmonic cycle (10) is the unique dynamically robust full-support equilibrium on (9). A tremble is absorbed by the phase alone: the level b is pinned by primitives, so no continuation changes the amplitude.

Proof. Free entry pins the locus: $b = \ln(\pi_0/\bar y)$ is set by primitives, so every full-support continuation lies on the original ellipse, and only the phase is free. From an interior displaced scale, Theorem 1 classifies the classical continuations in the transmitted direction as the harmonic path through that scale on the same orbit. The phase offset from the original path is $O(\varepsilon)$ where the original speed is bounded away from zero and $O(\sqrt{\varepsilon})$ at a turning point, and two paths on one orbit at phase offset θ differ by at most $2A|\sin(\theta/2)|$ at every date. The gap therefore vanishes with ε, and the cycle is robust. A rest path, and a pausing path during its pause, is not: an inward displacement forces $|\dot n_s^\varepsilon| = \sqrt{2h(n_s^\varepsilon)/\mu} > 0$ at once, so the continuation traverses the band within one period, an $O(A)$ departure for every $\varepsilon > 0$. ■

A tremble changes the phase of the cycle and nothing else. Free entry pins the level b, so a perturbed industry has no neighboring orbit to move to: the displaced scale sits on the same locus, slightly ahead of or behind the original clock. Rest and pausing paths instead rely on the industry being exactly still at a break-even scale. After an inward displacement, the flow condition sets the speed at once, and the continuation traverses the band. The selection uses only the pinned one-dimensional locus, so it holds on every landscape of Assumption 2. What the log-quadratic case adds is not stability but the frequency's independence of the level and the amplitude. The perturbations in Proposition 2 stay inside the full-support band. A downward displacement at n_- is therefore outside the refinement rather than an exception to it: by Proposition 3, such a displacement selects the withering path. The refinement is a selection among full-support paths, not global stability against every shock. The industry is thus most fragile at the lower break-even scale, and that margin is the theory's counterpart of the censored episodes in Section 4.3. Timing indifference pins the speed at each scale, and dynamic robustness picks the cycle out of the paths consistent with it. The smooth-schedule restriction of Theorem 1 is inessential for the selection, since the paths it rules out pause at a break-even scale and are not robust.

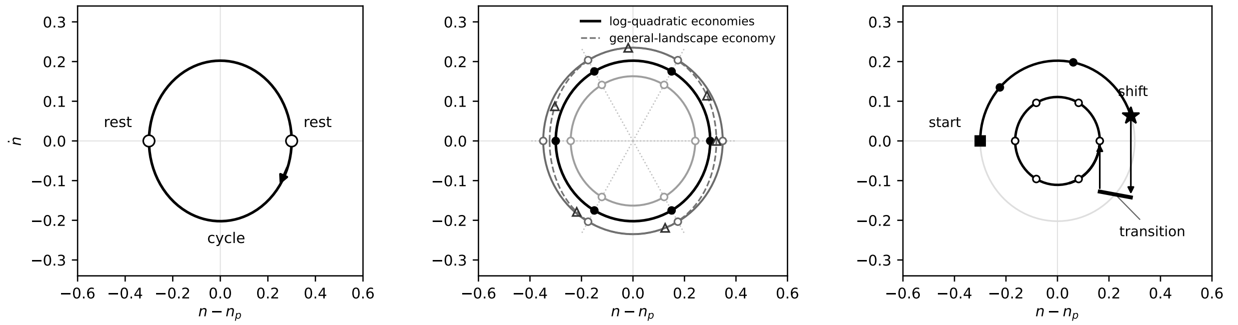
Frequency robustness over aggregate shocks Aggregate shocks enter as unanticipated events, deterministic surprises that are not priced before arrival. From the inherited scale, the continuation is an equilibrium of whatever environment prevails. The scale is the only inherited state, since the speed of adjustment is chosen through entry. A *transient* shock displaces the scale and leaves primitives unchanged; a *permanent* shock shifts productivity π_0, and with it the level b and the break-even band.

Proposition 3 (Frequency robustness over aggregate shocks)**.**
After an unanticipated shock at date τ^*, transient or permanent, every dynamically robust continuation enters the cycle of the prevailing environment, immediately when the inherited scale lies inside its band and after the unique no-entry transition when it lies above; a scale displaced below the band withers. Whenever the industry cycles, the frequency is $\omega = \sqrt{\varphi/\mu}$.

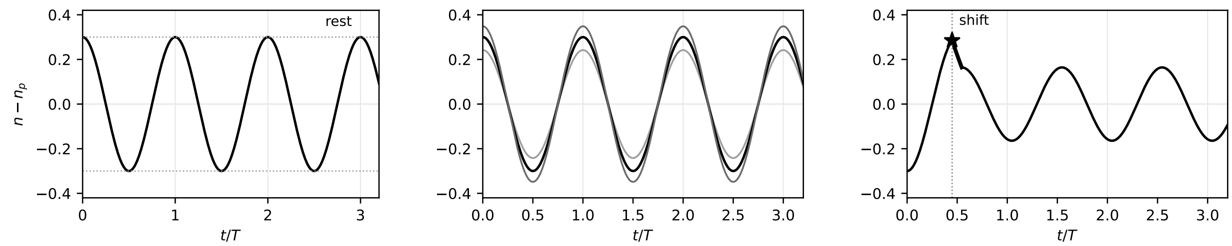
Proof. If the inherited scale n_{τ^*} lies strictly inside the prevailing band, the flow condition holds from τ^* on, Theorem 1 classifies the continuations from n_{τ^*}, and Proposition 2 selects the cycle through it, up to direction. If n_{τ^*} lies outside the prevailing band, profit falls short of the entry-cost annuity at every speed, so complementary slackness gives $e_t = 0$ and $n_t = n_{\tau^*} e^{-\delta(t-\tau^*)}$; the entry value stays below κ along the decay, consistent with zero entry. Exit only lowers the scale, so the two sides differ: from above the band the decay reaches n_+ in finite time, where the previous case applies; from below it moves away from n_-, profit stays below the annuity along the whole path, and the continuation withers. The frequency contains neither the level nor the phase, and a multiplicative productivity shift leaves the curvature of log profit unchanged. ∎

Transient shocks change the phase of the cycle, and permanent shocks change its amplitude. Neither changes the frequency. A transient shock re-enters the same orbit with a phase shift, since the environment is unchanged and the level-b locus again applies at the inherited scale. A permanent shift moves the industry to the orbit of the new level, and the transition respects the asymmetry between the two coordinates of the state: the scale is a stock and cannot jump, while the speed can, because the entry flow can. When the shift strands the scale beyond the new break-even scale, profit falls short of the entry cost at every speed, and entry stops at once. The scale then shrinks only through exit at rate δ. Entry resumes where

Figure 1: The cycle, its clock, and its response to shocks



(a) Baseline (b) Amplitude ⊥ frequency (c) Permanent productivity shift



(d) Time path of (a) (e) Time path of (b) (f) Time path of (c)

Notes: Top row: position and speed of adjustment $(n - n_p, \dot{n})$ with $\varphi = 1$ and $\mu = 2.2$, so $\omega = \sqrt{\varphi/\mu} \approx 0.67$; travel is clockwise. Bottom row: the corresponding time paths of the scale, with vertical gridlines one period apart. Panel (a): the locus (9) at level $b = 0.045$ ($A = 0.30$); open dots are the rest equilibria. Panel (b): equilibrium cycles at levels $0.65b$, b, and $1.35b$, each the cycle for a different break-even flow $\bar{y}$; markers at common dates, spaced one-sixth of a period apart, lie on common rays (dotted): every equilibrium cycle turns at the same ω, whatever its size. The dashed curve is the equilibrium cycle of a fourth economy whose log profit adds a quartic term: there the period depends on the orbit's size, so its markers (triangles) drift off the rays. Panel (c): from the square marker the industry travels the outer cycle; an unanticipated permanent drop in productivity at the starred date (Proposition 3) strands the inherited scale above the new break-even scale: entry pauses, the industry declines along the no-entry transition path (the exit rate is exaggerated for visibility), and the cycle resumes on the smaller orbit, the arrows marking the jumps in the entry flow. Dots one-sixth of a period apart before and after show the unchanged clock, visible in panel (f) as unchanged peak spacing.

profit again covers the cost, the new break-even scale, so the cycle continues from its turning point. The asymmetry has a second implication: a displacement below the lower break-even scale declines away from the band, so a severe enough adverse shock removes the industry rather than re-timing it. No selection is needed on the transition, where complementary slackness forces the path uniquely; dynamic robustness selects only where entry is positive. This transition path is the model's recession, a halt in entry and a gradual decline, after which the cycle runs again at the unchanged frequency. The historical windows in Section 4.4 show this pattern. Section 4.3 tests the response directly. Because every continuation runs at ω, the estimated spectral peaks of Section 4 are read as estimates of ω whatever the shock history. The shocks here are unanticipated; recurrent anticipated shifts would add a premium to the flow condition near the turning points and lie outside the proposition. The persistent-cost case is the conditional exercise of Online Appendix B. Figure 1 draws the geometry, in the phase plane and in time: the locus with its equilibria, equilibrium cycles of every size sharing the clock, and a permanent productivity shift resizing the cycle at an unchanged frequency.

Periodicity and isochrony The orbit is general; the constant frequency is not. The same argument runs on any single-peaked landscape of Assumption 2: the locus still confines the state to $[n_-, n_+]$, classical scheduling still forces reversal at the break-even scales, and every nonstationary classical full-support equilibrium is still a periodic orbit, now with period $T(\bar{y}) = \sqrt{2\mu}\int_{n_-}^{n_+}[\ln(\pi(n)/\bar{y})]^{-1/2}\mathrm{d}n$. What the log-quadratic case adds is amplitude independence: the period stops depending on the orbit's size, and the frequency collapses to $\sqrt{\varphi/\mu}$. That addition is less special than it looks. As $\bar{y} \uparrow \pi(n_p)$ the general period reproduces it locally, $2\pi/T(\bar{y}) \to \sqrt{-\pi''(n_p)/[\mu\pi(n_p)]}$, so every smooth peak is harmonic for small orbits. The log-quadratic landscape extends the small-orbit frequency to every amplitude, and it is not the only landscape that does. The period depends on the landscape only through the width of the well at each depth, so any landscape sharing its width function, however skewed, is isochronous at the same ω; Prediction 2 therefore tolerates landscape skewness. Online Appendix B.3 gives the statements.

Balanced growth The cycle extends to a growing industry once every primitive that sets the industry's scale grows with it, including the entry pipeline, so that congestion loads on adjustment relative to trend rather than on trend growth itself. Fix $g \in [0, \rho)$ and let productivity, the entry cost, and the maximal-profit scale grow at rate g: $\pi(n,t) = e^{gt}\,\hat{\pi}(n/n_{p,t})$ with $n_{p,t} = n_{p,0}e^{gt}$ and $\kappa_t = \kappa_0 e^{gt}$. The intensive forms are log-quadratic, $\hat{\pi}(u) = \pi_0 \exp[-\frac{\hat{\varphi}}{2}(\ln u)^2]$ and $\hat{\xi}(v) = e^{-\frac{\hat{\mu}}{2}v^2}$, with the friction applying to the log deviation $x_t \equiv \ln n_t - \ln n_{p,t}$.

Definition 2 (Balanced-growth timing-indifference equilibrium).
A balanced-growth full-support timing-indifference equilibrium *is a path $\{n_t\}_{t\geq 0}$ and an entry policy $e(\cdot)$ satisfying Definition 1 with the growing entry cost and pool: $V(\tau) = \kappa_\tau$ for every τ, and $n_t < M_t = M_0 e^{gt}$ for all t.*

The pool grows with the economy because the industry's scale does: along a growth path a fixed M would eventually bind.

Proposition 4 (Balanced-growth cycle)**.**
Let $b_g \equiv \ln \frac{\pi_0}{(r+\delta-g)\kappa_0} > 0$*. In any balanced-growth equilibrium the log deviation satisfies*

$$\frac{\hat{\varphi}}{2}\, x_t^2 + \frac{\hat{\mu}}{2}\, \dot{x}_t^2 = b_g, \tag{13}$$

and the classical full-support equilibria in x *are rest at* $x \equiv \pm\sqrt{2b_g/\hat{\varphi}}$ *and the cycle* $x_t = \tilde{A}\cos(\hat{\omega}t + \phi)$ *with* $\hat{\omega} = \sqrt{\hat{\varphi}/\hat{\mu}}$ *and* $\tilde{A} = \sqrt{2b_g/\hat{\varphi}}$*, provided* $\tilde{A}\,\hat{\omega} < \delta + g$ *and* $\tilde{A} < \ln(M_0/n_{p,0})$*.*

Proof. With κ_t growing at rate g, $V(\tau) = \kappa_\tau$ for all τ gives $\mathrm{d}V/\mathrm{d}\tau = g\kappa_\tau$, so differentiating (5) yields the flow condition $\pi(n_\tau, \tau)\,\hat{\xi}(\dot{x}_\tau) = (r + \delta - g)\kappa_\tau$. The common trend e^{gt} cancels, and taking logs gives (13). In $(x, \dot{x})$ this is the ellipse (9) verbatim, and the argument of Theorem 1 applies unchanged: rest at the two still points, or the harmonic cycle with amplitude fixed at $\tilde{A}$. For feasibility, $e_t = n_t(\dot{x}_t + g + \delta)$ is positive exactly when $\tilde{A}\hat{\omega} < \delta + g$, and $n_t < M_t$ exactly when $x_t < \ln(M_0/n_{p,0})$. ∎

The industry cycles harmonically around its growing peak, and rest becomes trend-following at a constant log gap. At $g = 0$ the specification reduces to the benchmark at leading order, with $\hat{\varphi} = \varphi n_p^2$ and $\hat{\mu} = \mu n_p^2$, so $\hat{\omega} = \omega = \sqrt{\varphi/\mu}$. The frequency is invariant to the growth rate, so cross-industry comparisons are undisturbed by industries sitting on different growth trends. Growth relaxes the entry-flow bound, since replacement and trend entry together absorb larger swings. The capacity-scaling primitive is essential: if the pipeline did not expand with the economy, trend growth would shift the friction's reference point, the ellipse in $(x, \dot{x})$ would center at $\dot{x} = -g$, and the still points would absorb the path rather than turn it. Growth neutrality therefore depends on the pipeline growing with its industry.

3.2 Theoretical predictions

Conditional on the cyclical equilibrium, the benchmark makes two predictions about frequencies. Neither prediction determines whether an industry belongs to the active or withering class. Conditional on full support, Proposition 2 selects the cyclical path; the empirical analysis asks whether its frequency implications appear in industries that fluctuate.

Prediction 1: industry specificity By (10) the frequency $\omega_i = \sqrt{\varphi_i/\mu_i}$ depends on the industry's own primitives and on nothing else, so industries with different profit landscapes or entry pipelines cycle at different rates. The cross-industry spread of $\ln \omega_i$ should therefore exceed what estimation error alone would produce. This prediction requires no proxy and no auxiliary mapping: it is a statement about the estimated frequencies themselves.

Prediction 2: monotonicity Taking logs in (10), independent industries satisfy

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i, \tag{14}$$

so a sharper log-profit peak raises frequency while a stiffer pipeline lowers it. The content is the *ordering*: holding the friction channel fixed, frequency should rise with profit curvature. Under general profits the period depends on the entire landscape at finite amplitude, and a skewed landscape makes the orbit spend unequal time on either side of n_p. Because the friction is symmetric in $\dot{n}$, the expansion and contraction legs remain exact time reversals of equal duration. Online Appendix B.3 and B.5 give the derivations; Online Appendix C reports numerical illustrations.

Taking Prediction 2 to data requires an additional mapping. Following the geographic-concentration concept of Ellison and Glaeser (1997), I construct the Ellison–Glaeser index γ_i for each industry from public County Business Patterns county and national files; Online Appendix D documents the construction, and every input is public. I treat the index as a correlate of profit curvature, not as a direct measure. If $\ln \varphi_i = a + \lambda \ln \gamma_i + v_i$ and $\ln \mu_i$ is orthogonal to $\ln \gamma_i$, the population slope of $\ln \omega_i$ on $\ln \gamma_i$ is $\lambda/2$. The model therefore motivates a positive ranking under those restrictions; it does not imply that the implemented coefficient must equal one-half.

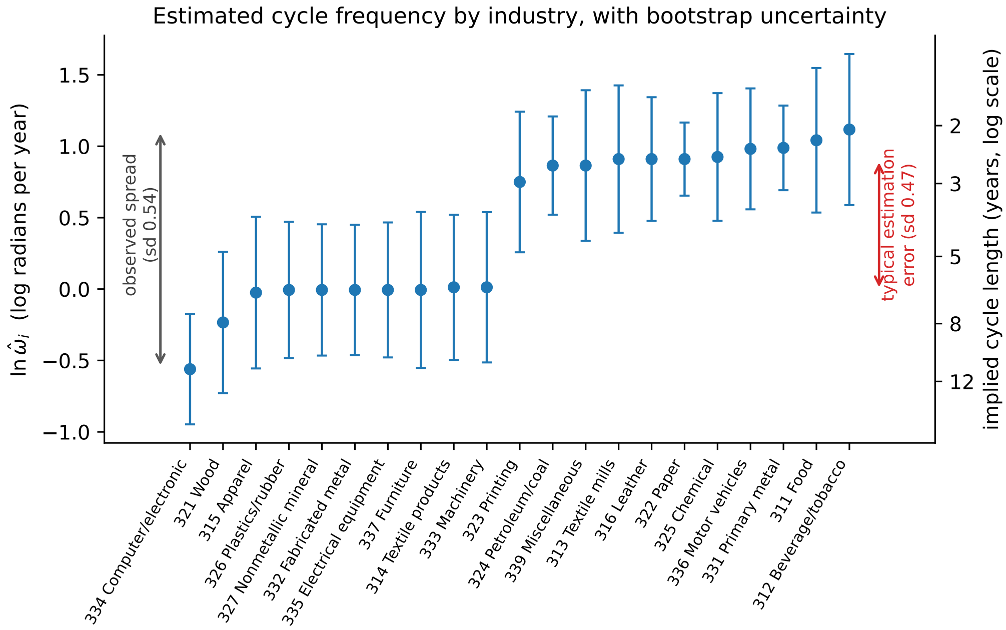
4 Empirical evidence

4.1 Industry specificity

Prediction 1 concerns the estimated frequencies themselves, so it can be assessed before any proxy is introduced. For each of the 21 industries I estimate $\ln \hat{\omega}_i$ with the five-taper rule of Section 4.2, and I measure how precisely that peak is estimated by simulating 200 series from the industry's own fitted AR(12) and re-estimating the peak on each.

Across the 21 industries the estimated cycle runs from 2.1 to 11.0 years (Figure 2). This spread arises under the common aggregate conditions the paper sets out to explain. The dispersion in log frequency, 0.54, exceeds the 0.47 estimation floor, in the direction Prediction 1 requires, although the formal common-frequency null is not rejected ($\chi^2(20) = 26.83$, $p = 0.14$). The test is weakly powered: at the estimated signal, a five-percent test rejects with probability of about 0.3. The data therefore do not sharply distinguish the predicted dispersion from a common frequency. Equivalently, roughly a quarter of the cross-industry variance in $\ln \hat{\omega}$ is signal, a reliability of 0.25 that Online Appendix D uses again. The margin over that floor is small, so the estimates place the industries in a broad pattern rather than a sharp ordering.

Figure 2: Estimated cycle frequency by industry, with sampling uncertainty



Notes: Each point is $\ln \hat{\omega}_i$ from the five-taper rule on monthly log production growth, 1972–2024. Bars are ± 1 sampling standard deviation from the industry's own parametric bootstrap. The arrows compare the spread of the estimates across industries with their typical sampling error.

That pattern is the phenomenon the model is built to explain.

4.2 Cross-industry ranking

Table 1: Concentration and estimated industry frequency

	Annual establishments	Production
Source	BDS	FRED IP
Period	1978–2022	1972–2024
Industries	15	21
Peak estimator	Burg AR(6)	DPSS, $K = 5$
$\hat{\beta}_\gamma$	0.18	0.06
HC1 s.e.	(0.09)	(0.08)

Notes: Each column regresses log estimated frequency on the log Ellison–Glaeser index built from public County Business Patterns data (state level, 2021; Online Appendix D). Production frequencies use the largest five-taper peak in the fixed 2–12-year band. BDS frequencies use log-differenced establishment counts and Burg AR(6) peaks in the same band; three of its 15 estimates lie at a band boundary, and excluding them gives 0.18 (HC1 0.07). HC1 standard errors treat estimated peak locations as observed.

The model's state is the establishment count, so the most direct reading of Prediction 2 ranks establishment-based frequencies. Annual BDS establishment frequencies rise with concentration. The universe here is 15 of the 21 production industries, the ones whose NAICS-3 establishment counts can be built from the public BDS files (Online Appendix D).

Across these 15 industries, the slope of log frequency on the log Ellison–Glaeser index is 0.18 (HC1 s.e. 0.09). Excluding three estimates at the band boundary leaves the slope at 0.18 and tightens the standard error to 0.07, distinguishable from zero at conventional levels. The slope is positive in all four variants of the index, county or state, 2016 or 2021, with t-statistics from 1.8 to 2.4, and from 2.2 to 2.8 once the boundary estimates are excluded.[4] The result does not depend on treating the estimated peaks as data. A parametric bootstrap refits each industry's annual dynamics, re-estimates the peaks, and re-runs the regression. The bootstrap slopes center at 0.12 rather than 0.18, because re-estimated peaks attenuate the slope toward zero. Their standard deviation equals the HC1 s.e., and the slope is positive in 91 percent of draws (Online Appendix D).

Production widens the sample to all 21 NAICS-3 industries under one pre-specified rule, but it is not the model's state. For each industry I take the aggregate FRED production index over 1972–2024 (series `IPGxxxS`), estimate a five-taper Thomson DPSS spectrum of monthly log production growth, and select the largest peak in a fixed 2–12-year band. The production slope is positive, 0.06 (HC1 s.e. 0.08), but not distinguishable from zero. It is not a recession-timing artifact: purging a common factor or NBER recession dummies from the growth series before peak estimation moves it to 0.08 and 0.07. Table 1 collects both designs.

Annual BDS establishment log frequencies correlate 0.32 with production log frequencies under the fixed-band protocol ($p = 0.24$). A quarterly establishment panel is uninformative for this design: quarterly net entry is small relative to the stock and a common component absorbs most of its variation, so it identifies no industry-specific frequencies; Online Appendix D reports it.

Screened 15-industry designs under the legacy archived score give larger but peak-selection-sensitive estimates. That estimator panel, the archived score itself, and a historical-instrument check of concentration endogeneity are in the Online Appendix. The archived score correlates 0.73–0.81 with the built index. Together these exercises support a model-inspired ranking without identifying φ_i, μ_i, or the concentration-to-curvature mapping.

4.3 Depth and recovery

The equilibrium response makes a parameter-free prediction about recessions. Under Proposition 3, a transient shock displaces the scale, and the recovery is travel along the unchanged orbit back to the peak. A fall of size $\varDelta$ therefore takes $\arccos(1 - \varDelta/A)/\omega$ to reverse, and the slope of log recovery time on log depth is one-half at leading order, independent of the amplitude, the frequency, and the primitives. The competing exponents are zero and one.

[4] Because the rule assigns a peak to every series, the regressions rank protocol-specific frequencies rather than certifying a periodic component in each industry.

Zero arises from mapping deeper recessions into larger orbits, a mapping the model does not make. One arises from recovery that closes the gap at a constant rate, as trend growth does. Direction is free in Proposition 3, and the design is direction-proof: the measured trough is the realized minimum, so a continuation that falls first registers the full swing as its depth and the half period as its recovery, the endpoint of the same curve.

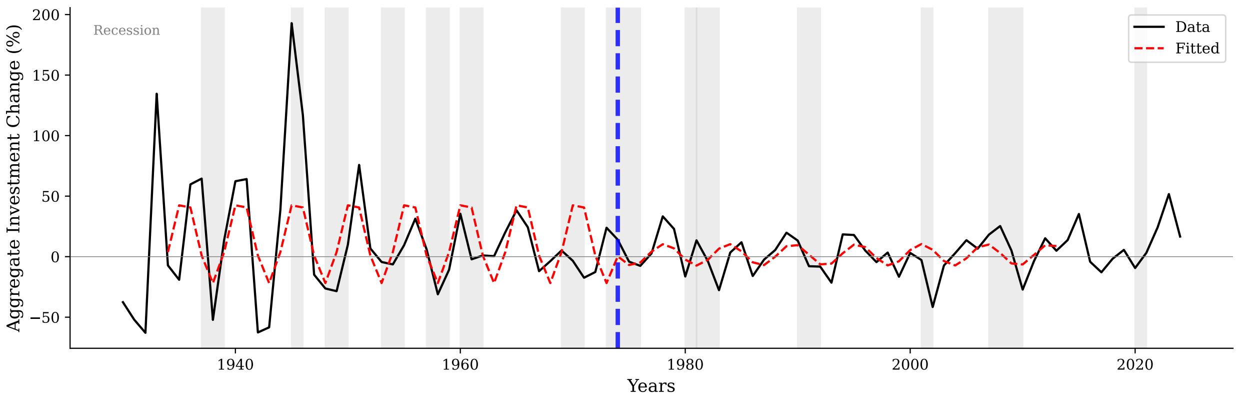
Across 85 recovering industry–recession episodes in monthly production, with industry fixed effects and standard errors clustered by recession, the slope is 0.469 (s.e. 0.179): zero and one are rejected, one-half is not ($t = -0.17$). The test is best defined in the fifteen industries with growing production, where a prior peak is a recoverable target. The slope there is 0.492 (s.e. 0.190), on the predicted value, and both alternatives remain rejected at the five-percent level even on a t_6 reference. The forty-nine episodes that never regain their prior peak within five years are consistent with the non-recovering regime, permanent downshifts or displacements below the lower break-even scale, rather than with slow recoveries. They are deeper, and industry trend growth predicts censoring given depth. Censoring concentrates in declining industries (54 against 29 percent). Pooling the two regimes into one censored regression returns exponent one, which is what conflating them produces. The decline itself carries no depth dependence (0.146, s.e. 0.177): the fall reflects the shock, and the recovery reflects the cycle. The exponent identifies recovery as traversal past a smooth peak, a signature any smooth limit-cycle model shares. The discriminating content is the joint rejection of depth independence and constant-rate recovery together with the censoring pattern. Production is not the model's establishment state and seven clusters bound the precision; Online Appendix D reports the design and all variants.

4.4 Historical echoes

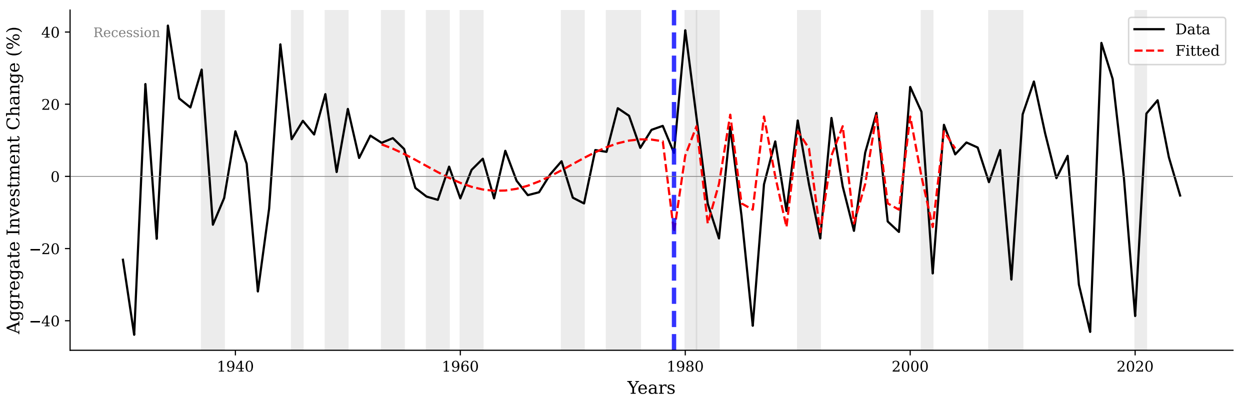
Two long historical windows make the paper's object visible in the raw series. From 1934 to 1973, manufacturing investment in structures rose and fell on a five-year rhythm (Figure 3a), through four decades that contain the Depression recovery, the war economy, demobilization, and the postwar boom. Mining and oil investment after the 1979 oil shock cycled faster, alternating every 3.25 years across a quarter century that spans the 1986 price collapse and the Gulf War (Figure 3b). Aggregate conditions changed repeatedly inside each window; the rhythms persisted, and they differ across the two industries. Frequency appears to be a property of the industry rather than of the era.

The industries, windows, and specification were chosen after inspecting the episodes, and the series are investment flows rather than the model's establishment stock. Figure 3 is therefore an illustration rather than a test; its value is to make the differing rhythms visible.

Figure 3: Fitted periodic components in two historical investment windows



(a) Manufacturing structures



(b) Mining exploration, shafts, and wells

Notes: Annual growth in private fixed investment from BEA NIPA Table 5.4.1. Solid line: data; red dashed line: fitted cosine; gray bands: NBER recessions. Blue dashed lines separate each event window from its equal-length comparison window. Within each window, the frequency maximizes the demeaned Fourier periodogram and the cosine is fitted by OLS. The event-window periods are $T = 5.0$ years for manufacturing in 1934–1973 and $T = 3.25$ years for mining/oil in 1979–2004. These components do not distinguish timing search from vintage replacement, structural breaks, or other propagation mechanisms.

Online Appendix D reports the window definitions and the fitting procedure.

5 Pigouvian taxation

The normative benchmark is the planner's resource problem. The planner chooses gross entry to maximize the present value of the industry's contribution to household consumption, effective profit net of entry spending:

$$\int_0^\infty e^{-\rho t}\big[n_t\pi(n_t)\xi(\dot n_t) - \kappa e_t\big]\,dt$$

subject to $\dot n_t = e_t - \delta n_t$. Locally, a rest optimum n^* satisfies

$$\pi(n^*) + n^*\pi'(n^*) = (\rho+\delta)\kappa. \tag{15}$$

Free entry rests where an entrant's own profit just covers the cost, $\pi(n) = (\rho+\delta)\kappa$; the planner's condition carries the extra term $n^*\pi'(n^*)$, the profit a marginal entrant adds to, or removes from, incumbents through the scale channel. Free entry ignores this externality. Above the profit peak the term is negative, an extra firm crowding the market, so free entry brings in too many firms and the planner would tax entry; below the peak the sign reverses. Proposition 5 prices the externality state by state, along the whole path rather than only at rest, so that private free entry reproduces the planner's allocation.[5]

Proposition 5 (State-dependent Pigouvian wedge)**.**
Along an interior planner path, the state-dependent rebated entry wedge

$$\tau_t = -n_t\pi(n_t)\xi'(\dot n_t) + \int_t^\infty e^{-(\rho+\delta)(s-t)}\left[\delta n_s\pi(n_s)\xi'(\dot n_s) - n_s\pi'(n_s)\xi(\dot n_s)\right]\mathrm{d}s \tag{16}$$

decentralizes that interior planner allocation. At a rest point,

$$\tau^* = -\frac{n^*\pi'(n^*)}{\rho+\delta}.$$

It is positive when the optimum lies on the crowded side of the peak, $n^* > n_p$.

Proof. Take entry as the control, with current-value Hamiltonian

$$\mathcal{H} = n\,\pi(n)\,\xi(e-\delta n) - \kappa e + \lambda\,(e-\delta n), \qquad \dot n = e - \delta n.$$

The entry margin $\partial\mathcal{H}/\partial e = 0$ gives

$$\lambda_t = \kappa - n_t\pi(n_t)\,\xi'(\dot n_t), \tag{17}$$

and $\dot\lambda = \rho\lambda - \partial\mathcal{H}/\partial n$ gives the costate equation

$$\dot\lambda_t = (\rho+\delta)\lambda_t - \big[\pi(n_t) + n_t\pi'(n_t)\big]\xi(\dot n_t) + \delta\, n_t\pi(n_t)\,\xi'(\dot n_t),$$

[5] The planner's optimum is a rest point to leading order, which requires concavity of $n\pi(n)\xi(\dot n)$ rather than of π alone. Online Appendix E gives the derivation and the flat-tax implementation.

which, under $e^{-(\rho+\delta)T}\lambda_T \to 0$, integrates forward to

$$\lambda_t = \int_t^\infty e^{-(\rho+\delta)(s-t)} \Big\{ \big[\pi(n_s) + n_s\pi'(n_s)\big]\xi(\dot n_s) - \delta\, n_s\pi(n_s)\,\xi'(\dot n_s) \Big\} \mathrm{d}s. \tag{18}$$

By (17) the first term of (16) is $\lambda_t - \kappa$, so substituting (18) collects the two integrals into one. Their integrands sum to

$$\big\{(\pi + n\pi')\xi - \delta n\pi\xi'\big\} + \big\{\delta n\pi\xi' - n\pi'\xi\big\} = \pi(n_s)\,\xi(\dot n_s),$$

the scale and replacement terms cancelling, so that $\tau_t = \int_t^\infty e^{-(\rho+\delta)(s-t)}\pi(n_s)\xi(\dot n_s)\mathrm{d}s - \kappa = V_t - \kappa$, using $r = \rho$. Hence $V_t = \kappa + \tau_t$: the planner's path satisfies private free entry under (16). At a rest point $\xi'(0) = 0$ removes the first term and the integral reduces to $-n^*\pi'(n^*)/(\rho+\delta)$. ■

The wedge splits into the two channels through which an entrant affects others. The first term, $-n_t\pi(n_t)\xi'(\dot n_t)$, prices the adjustment the entrant adds at the moment of entry: since $\xi' = -\mu\dot n\xi$, it taxes entry while the industry expands and subsidizes it while it contracts. The integral capitalizes the scale externality $-n\pi'$ imposed on incumbents over the entrant's life, net of its interaction with replacement entry. At a rest point the first term vanishes, leaving the static Pigouvian tax $\tau^* = -n^*\pi'(n^*)/(\rho+\delta)$.

Implementation The timing component conditions only on observables. Its arguments are the industry's scale and net adjustment rate, which establishment counts measure directly. Its sign requires no estimate of the primitives: it taxes entry while the industry expands and subsidizes it while the industry contracts. The scale component does not share this structure: it is forward-looking, capitalizing $-n\pi'$ over the entrant's life. It can offset the timing sign where the two channels oppose, as on the expansion leg below the profit peak, where entry congests adjustment but raises incumbents' profits. The offset is largest just after each turning point, where the timing term is small; there the total wedge's sign, like its level, requires the primitives. Existing instruments already price entry timing in the timing component's form: volume-based permitting and licensing fees charge entrants more when applications surge, and countercyclical capital buffers charge bank expansion in booms and release the charge in contractions. In the semiconductor illustration, an entry subsidy granted during a rush, as capacity-expansion programs typically are, amplifies the congestion externality the timing component prices; the same subsidy shifted to the contraction phase works with it. Setting the wedge's level, rather than the timing component's sign, requires the primitives (φ, μ), the measurement agenda of Section 6.

Magnitudes Two back-of-envelope numbers give the scale of the wedge. At a rest optimum, dividing the planner condition into the wedge gives $\tau^*/\kappa = \varepsilon/(1-\varepsilon)$, where $\varepsilon \equiv -\partial \ln \pi/\partial \ln n$ is the elasticity of per-establishment profit with respect to industry scale at n^*. If a one-percent industry expansion lowers per-establishment profit by 0.3 percent, the level charge is roughly forty percent of the entry cost. Along the cycle, the flow condition turns the timing component into $-n\pi\xi' = (r+\delta)\kappa\,\mu n\dot{n}$. At the height of the rush $\mu\dot{n}^2 = 2b$, so the charge relative to the entry cost is $2b(r+\delta)/g_{\max}$, where b is the log profit loss from congestion there, approximately the share dissipated when small, and $g_{\max}$ is the peak growth rate of the industry's scale. Suppose congestion dissipates five percent of profit at the peak of the rush, so $b = -\ln(0.95) \approx 0.05$, the scale swings six percent per year, and $r + \delta = 0.12$. The swing sits inside the amplitude range of the feasibility diagnostic in Online Appendix B.4. The timing charge is then about a fifth of the entry cost at the height of the rush, with the matching subsidy at the trough of the retreat. Both inputs are elasticities and shares rather than structural parameters, and the numbers are illustrative.

Multiplicity under the flat tax The separate cycle-eliminating tax

$$\tau^{\text{collapse}} = \frac{\pi(n_p)}{r+\delta} - \kappa$$

raises the normalized private entry wedge to one. Under full-support timing indifference, the payoff locus then shrinks to the stationary point $(n_p, 0)$ because $\pi(n) \le \pi(n_p)$ and $\xi(\dot{n}) \le 1$, with joint equality only there. Under ordinary free-entry complementarity, however, the same tax also supports extinction from any inherited n_0: set $e_t = 0$ and $n_t = n_0 e^{-\delta t}$. Because $\pi(n_t)\xi(-\delta n_t) \le \pi(n_p)$, entry value never exceeds the taxed cost $\kappa + \tau^{\text{collapse}} = \pi(n_p)/(r+\delta)$. Thus the same flat tax supports both parking at n_p and extinction from any inherited scale: in practical terms, a licensing moratorium calibrated to stop the cycle also supports the industry's disappearance. It generally does not implement n^* and is not the state-dependent Pigouvian wedge in Proposition 5.

6 Concluding remarks

This paper studies equilibrium when entrants search over dates and every active date must offer the same value. The model admits withering and active rest and, when the full-support feasibility bound holds, cycling. Within the classical full-support cyclical class, every nonstationary equilibrium is the same orbit up to phase and therefore has the same period.

The general profit curve fixes that period through an exact integral, and log-quadratic profits collapse it to the closed form $\omega = \sqrt{\varphi/\mu}$, a profit-heterogeneity channel across industries. Taken to U.S. manufacturing with a concentration index built from public data, the establishment-based ranking is positive and distinguishable from zero and the production-based ranking is positive but imprecise. Recovery times rise with depth at the predicted exponent of one-half, rejecting constant-rate recovery. The historical echoes make the frequency object visible.

Three boundaries matter for interpretation. First, the symmetric friction concerns net adjustment of an inherited industry. It misses heavy simultaneous gross entry and exit, so a separate gross-entry and exit structure need not share the same frequency formula. Second, geographic concentration is not measured profit curvature and may covary with the timing friction. Third, the historical echoes are selected illustrations rather than inference.

The empirical next step is to measure both primitives independently: profit curvature from profits, markups, or survival as industry scale changes, and timing friction from permitting, construction, equipment-delivery, or financing delays. On the theory side, endogenous exit and separate gross-entry and exit pipelines would clarify the range over which net adjustment is a sufficient state. Nonlinear household welfare and production networks are subsequent extensions.

Equal private values across dates do not price the scale and adjustment effects on incumbents, which yields the state-dependent Pigouvian wedge. But the simpler flat tax likewise supports both parking at n_p and extinction. Timing search accounts for industry frequency with a single closed-form expression in two primitives, $\omega = \sqrt{\varphi/\mu}$.

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